

Taking the Framework Apart: What Each Piece Does, and What Generalizes

Working note (for me, before the next audience)

October 1, 2026

Purpose. This note dismantles the model into its working parts, switches each one off in turn, and records what happens to (i) the level of welfare losses, (ii) the optimal exchange-rate response, and (iii) the regime ranking. It answers the questions raised after the July 22 presentation (Christian: “how does the optimal regime change with and without the network?”; Adam: “what is the optimal weight on the exchange rate, and how does it change with the network?”) and ends with a crib sheet of audience questions. All numbers come from the 3-sector Chile calibration (the only one for which a joint rule optimization is cheap); losses are in units of 10^{-4} . Code: `code/opt_rule_point.m` (+ `_hipi`, `_slice`, `_peg` variants), `code/analysis_opt_rule.py`, `code/network_anatomy.py`; raw output in `results/opt_rule_*.csv`.

1 The one-paragraph version

Four things happen in the model, and each can be isolated. (1) A *production network* propagates cost shocks between sectors. (2) *Heterogeneous price stickiness* turns those cost shocks into markup dispersion in the sticky sectors, which is what welfare penalizes. (3) The *exchange rate* is a second cost-push channel, entering every sector’s Phillips curve through $\Gamma \Delta e_t$, with $\Gamma \propto \tilde{A} \Omega^F$: direct imports *plus* imports inherited from suppliers. (4) A *financial shock* (risk premium in UIP) moves Δe_t or the interest rate for reasons unrelated to the real economy, and the policy rule decides which of the two moves. The regime comparison is mostly a statement about (4) and about how aggressively the rule fights inflation; the network and stickiness (1)–(3) decide how much an explicit exchange-rate response adds on top.

2 The pieces, and what each contributes

Piece	Where it lives	What it does to the results
Domestic IO network Ω^H (density ρ)	$\mathcal{V}, \mathcal{B}, \Gamma$, Domar weights λ_D	Raises the optimized loss ($3.7 \rightarrow 6.3 \rightarrow 9.5$ for $\rho = 0, 1, 1.5$ at baseline shocks); raises the value of an FX response to <i>real</i> shocks from 0.4% to 12%
Heterogeneous stickiness $\hat{\delta}_i$	welfare weights κ_i , DC index ϕ^{DC}	Without it the FX term is worth less ($5.9\% \rightarrow 4.2\%$ of loss at $\rho = 1$; $3.3\% \rightarrow 1.6\%$ at $\rho = 0$)
Imported inputs Ω^F	Γ	Without direct imported inputs the FX term is worth 3.6% instead of 5.9% (consumption-side and export channels remain)
Risk-premium shock ε^{RP}	UIP	Makes Peg catastrophic; gives the FX term a network- <i>independent</i> value ($\approx 11\%$ when doubled, at every ρ)
Debt elasticity Ψ	UIP closing device	Second-order for the rule (5.7% vs 5.2% for $\Psi = 0.005$ vs 0.1)
Policy aggressiveness ϕ_π	rule	<i>The</i> dominant lever: raising ϕ_π from 1.5 to ≈ 20 cuts Managed's loss by 38% and Float's by 74%

Table 1: Anatomy of the model: effect of each ingredient (3-sector Chile, $\phi_\pi = 20$, $\phi_y = 1$ slice unless noted).

3 Optimal exchange-rate response with and without the network

3.1 Setup

I solve the model once per structural scenario and then re-solve it on a grid of the rule $\log(I/I^*) = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \log S_t$ (451 points per scenario: $\phi_\pi \in [1.2, 300]$, $\phi_y \in [0, 2]$, $\phi_s \in [0, 3]$), computing welfare from analytic second moments with the model's own Domar weights (which change with ρ). The baseline cell reproduces the headline managed-float loss (10.17). Scenarios: network density $\rho \in \{0, 0.5, 1, 1.5\}$ crossed with risk-premium shock size $\{0, 0.5, 1, 2\} \times$ baseline, plus ablations.

3.2 Result 1: the optimal rule is close to strict targeting of the divine-coincidence index

In every scenario the loss is minimized at a very hard response to the DC-index inflation ($\phi_\pi \approx 20$ –50) and $\phi_y \approx 1$; the loss surface is flat beyond $\phi_\pi \approx 20$. This is the Rubbo logic surviving in the open economy: targeting π^{DC} removes the productivity cost-push, and what is left is the exchange-rate channel $\phi^{DC\top} \Gamma \Delta e_t$ (the divine-coincidence breakdown result of the paper). The exchange-rate term in the rule is a *correction for that leftover*, not the main event.

3.3 Result 2: how much the FX term is worth depends on the network, and on which shock

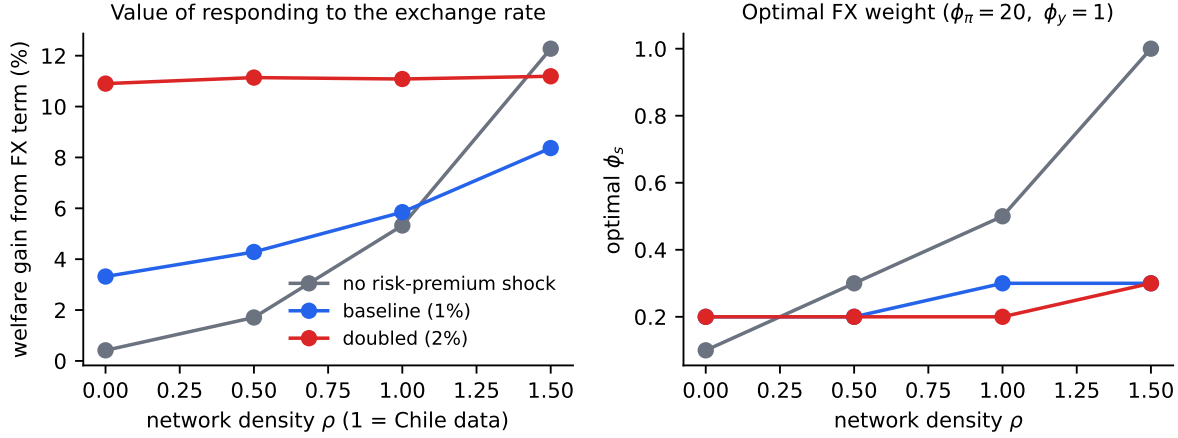


Figure 1: Value of adding $\phi_s > 0$ to the near-optimal Taylor rule, and the optimal ϕ_s , by network density and risk-premium shock size ($\phi_\pi = 20, \phi_y = 1$).

ρ	Value of FX term (% of loss)				Optimal ϕ_s			
	0	0.5	1	1.5	0	0.5	1	1.5
No risk-premium shock	0.4	1.7	5.3	12.3	0.1	0.3	0.5	1.0
Baseline (1%)	3.3	4.3	5.9	8.4	0.2	0.2	0.3	0.3
Doubled (2%)	10.9	11.1	11.1	11.2	0.2	0.2	0.2	0.3
Optimized loss, baseline shocks	3.68	4.58	6.26	9.50				

Table 2: Network $\times$ risk-premium grid at the near-optimal Taylor rule ($\phi_\pi = 20, \phi_y = 1$). “Value” is the loss reduction from choosing ϕ_s^* rather than $\phi_s = 0$. Plateau: the loss is within 1% of its minimum over a range of ϕ_s (e.g. 0.2–0.3 at $\rho = 1$), so ϕ_s^* is poorly identified when the gain is small.

Two separable stories:

1. **Against real shocks, the network is what makes an FX response worthwhile.** With the risk-premium shock off, the FX term is nearly worthless without a network (0.4%) and worth 12% at $\rho = 1.5$; the optimal weight rises $0.1 \rightarrow 1.0$. Mechanism: the network lets import-cost shocks reach sticky, import-light sectors (Services) through suppliers, so $\phi^{DC^\top} \Gamma$ grows with ρ (the DC-residual ratio $\phi^{DC^\top} \Gamma / \phi^{DC^\top} \mathcal{B}$ rises from 0.15 to 0.21 over $\rho = 0 \rightarrow 1.5$, [results/network_anatomy.csv](#)) and also because every cost shock is more costly when it propagates.
2. **Against the financial shock, the network is irrelevant.** The risk-premium shock moves the exchange rate directly; the value of leaning against it is $\approx 11\%$ at every ρ when the shock is doubled, with $\phi_s^* \approx 0.2$. A policy-maker facing mainly UIP shocks should lean against the exchange rate whether or not the economy has a production network.

So the honest answer to “does the optimal FX response depend on the network?” is: *yes, for real shocks; no, for financial shocks; and in the baseline calibration (equal 1% shocks) the gain from the FX term is modest, 3–8% of loss.*

3.4 Result 3: the regime ranking is largely a statement about rule aggressiveness

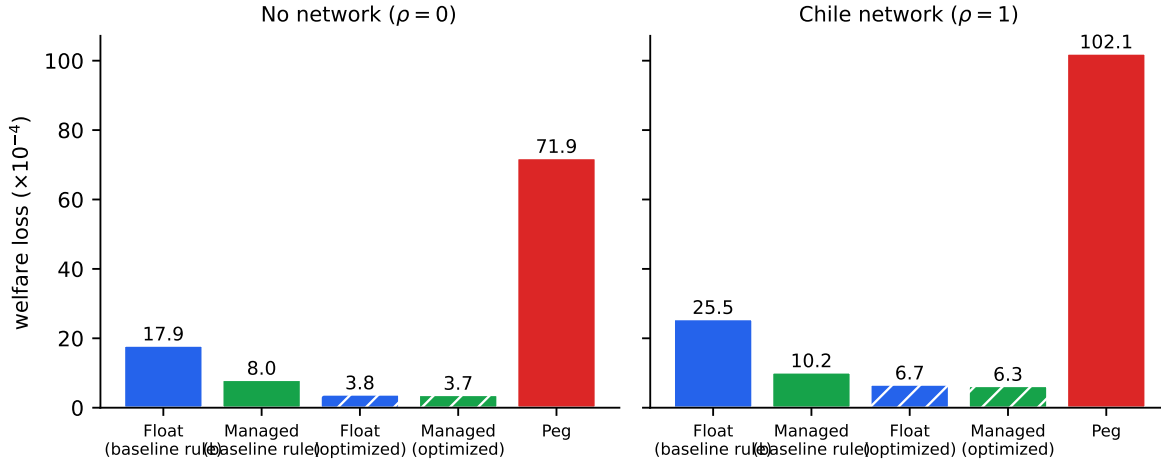


Figure 2: Welfare loss by regime with the paper’s baseline rule ($\phi_\pi = 1.5, \phi_y = 0.5$) versus optimized rules, baseline shocks.

	Float (base)	Managed (base)	Float (opt.)	Managed (opt.)	Peg
$\rho = 0$ (no network)	17.9	8.0	3.80	3.68	71.9
$\rho = 1$ (Chile)	25.5	10.2	6.65	6.26	102.1

Table 3: Regime comparison at baseline vs. optimized rules (baseline shocks).

The headline “Managed beats Float by 2.5 $\times$ ” compares a weak float rule ($\phi_\pi = 1.5$) with a managed rule whose FX term partly *substitutes for* the missing inflation response. Once both are optimized, Managed beats Float by only 3–6%. What does survive: (a) Peg is the clear loser, 16–20 $\times$ worse than an optimized rule, with or without the network, and even with the risk-premium shock off (Peg 25.2 vs. 6.1 at $\rho = 1$), because giving up the policy rate costs a lot against *real* shocks too; (b) the loss of every regime rises with network density, Peg fastest. **The paper’s regime comparison should be presented as “Peg vs. an optimized rule, with a small additional gain from an FX term”, not as “Managed beats Float by a wide margin”.**

3.5 Result 4: where the import exposure sits relative to stickiness

Christian’s example (rigid “real economy” sector buying from a flexible import-export sector) says the *location* of exposure relative to stickiness matters. Permuting the three import shares $\{0.0767, 0.1945, 0.0704\}$ across sectors:

Import-heavy sector	Resource (flexible)	Manufacturing (data)	Services (sticky)
Loss, $\rho = 1$, real shocks only	5.75	6.09	6.79
Value of FX term (%)	4.9	5.3	7.5

Putting the import-heavy role on the stickiest sector raises the loss by about 12% and raises the value of an FX response by about two points; at $\rho = 0$ the same permutation moves the loss by at most 0.6 (the effect needs the network to transmit exposure). Effects are real but modest in this calibration, because the calibrated import shares differ by at most a factor of three and the stickiest sector (Services) is also the one with the lowest direct import share.

4 What generalizes, and under what conditions

Claim	Holds when	Fails or is weak when
DC breaks under an exchange-rate channel (theory)	$\Omega^F \not\propto \alpha$ and some pass-through ($\Gamma \neq 0$)	Pass-through is zero (full dominant-currency pricing of imports) or $\Omega^F \propto \alpha$
Network raises the value of an FX response to real shocks	Stickiness is heterogeneous and import exposure is inherited through suppliers	Stickiness is uniform: gain falls to 1.6–4.2%
Peg is by far the worst regime	Any plausible rule can be chosen for the alternative; shocks can move the policy rate	Not tested: pegs with capital controls, or economies where I^* shocks dominate
Risk-premium shocks dominate Peg's loss	σ_{RP} is comparable to other shocks	$\sigma_{RP} \rightarrow 0$: Peg is still 4× an optimized rule but roughly equals a weak float
Managed > Float by a wide margin	The float rule is unaggressive ($\phi_\pi \approx 1.5$)	Both rules optimized: 3–6% only

5 Crib sheet: questions an audience will ask

“Why does your managed float win?”

Only against a weak plain float. With the inflation response optimized ($\phi_\pi \approx 20$) the gap closes to 3–6%. The robust claim is that an explicit exchange-rate term never hurts and helps most when the network is dense or UIP shocks are large.

“What is the optimal weight on the exchange rate?”

Small: $\phi_s^* \approx 0.2$ – 0.3 at baseline, rising to ≈ 1 against real shocks in dense networks. Because the loss is flat in ϕ_s near the optimum (within 1% over 0.2–0.3) the weight is weakly identified; the *value* of the term is the better-identified object.

“Why does the network matter for the FX weight?”

Because the DC index cannot remove the exchange-rate cost-push, and its size is $\phi^{DC\top} \Gamma$, with $\Gamma \propto \tilde{A} \Omega^F$ including inherited imports. A denser network raises the inherited part ($\approx 27\%$ of Services' import exposure at $\rho = 0$ vs. 41% at $\rho = 1$).

“Is a peg ever good here?”

Not in this model: it forfeits the policy rate, and the risk-premium shock then goes straight to the output gap. Without the shock, the peg is still $4\times$ worse than an optimized rule. Caveats: no capital controls, no credibility benefit, no fiscal side.

“Is the model over-identified (Phillips curve + UIP + law of one price)?”

No: Phillips curves price sectors, UIP prices the currency, LOP prices imports. Together they determine inflation, the exchange rate and the rate, given the rule.

“What if import prices are invoiced in dollars?”

Not modeled (producer-currency pricing). Dominant-currency pricing would weaken Γ for the domestic-currency exchange rate and shift weight to the dollar rate; a natural extension.

“How big is each ingredient?”

See the anatomy table: ϕ_π dominates the level; the risk-premium shock dominates the Peg loss; the network and stickiness heterogeneity decide the value of the FX term.

“Does 12/33/81 sectors change this?”

The ranking is stable; Peg’s loss is invariant to resolution while Float’s falls (diversification of sectoral shocks). The optimal-rule exercise has been run at 3 sectors only.

“Which parameter would you estimate first?”

The risk-premium shock’s volatility: it sets Peg’s loss and the network-independent part of the FX value. Then sectoral shock volatilities, which set the level of Float and Managed losses.

6 Open issues in the framework itself

- The optimal-rule exercise uses one 3-sector calibration and equal shock volatilities; the multi-country, many-sector version is the obvious next run (use the same approach with simulated moments).
- Welfare is the second-order approximation of the closed-economy form extended by the cross-sector term; the cross-sector term is not included in the optimization grid (it adds 3–8% and does not alter rankings in the baseline).
- Rule class is restricted to Taylor-type rules in $(\pi^{DC}, \tilde{y}, \log S)$; a fully optimal (Ramsey) policy is not computed. The finding that $\phi_\pi \rightarrow \text{large}$ is optimal suggests Ramsey would deliver strict DC-index targeting plus a correction for $\Gamma \Delta e_t$.
- The optimized rule requires the central bank to observe π^{DC} , the Domar-and-stickiness-weighted index; a CPI-based rule is worse in the earlier robustness exercise.